

Variational Quantum Algorithms

Motivation

- Our goal is to understand and conceptualise the physical world around us.
- Quantum mechanics gives us well-tested laws for microscopic degrees of freedom.

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle$$

- Knowing the laws does not make the collective behaviour easy to predict. “More is different” — Philip Anderson

The quantum many-body problem

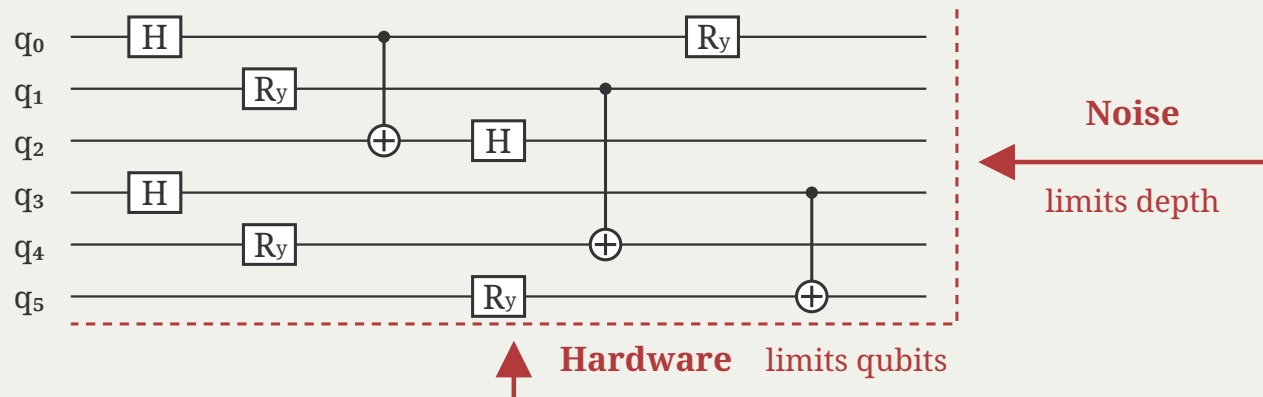
- N sites, each with a local dimension d :

$$|\psi\rangle = \sum_{\sigma_1, \dots, \sigma_N} \psi_{\sigma_1, \dots, \sigma_N} |\sigma_1, \dots, \sigma_N\rangle,$$

- A generic wavefunction has d^N expansion coefficients.
- Take $d = 2$. Write each amplitude on paper and place it in its own 1 cm^3 box.
- At roughly $N = 287$, you have enough boxes to fill the volume of the observable universe.
- How do we proceed?

Could a quantum computer help?

- N qubits can represent a state in a 2^N -dimensional Hilbert space.
- Can we efficiently prepare a desired state of interest?
 - We can try!
- What are the current limitations?



The variational approach

1. Choose an ansatz

A parameterised trial state $|\psi(\boldsymbol{\theta})\rangle$.

2. Define an objective

For a ground state, minimise the energy:

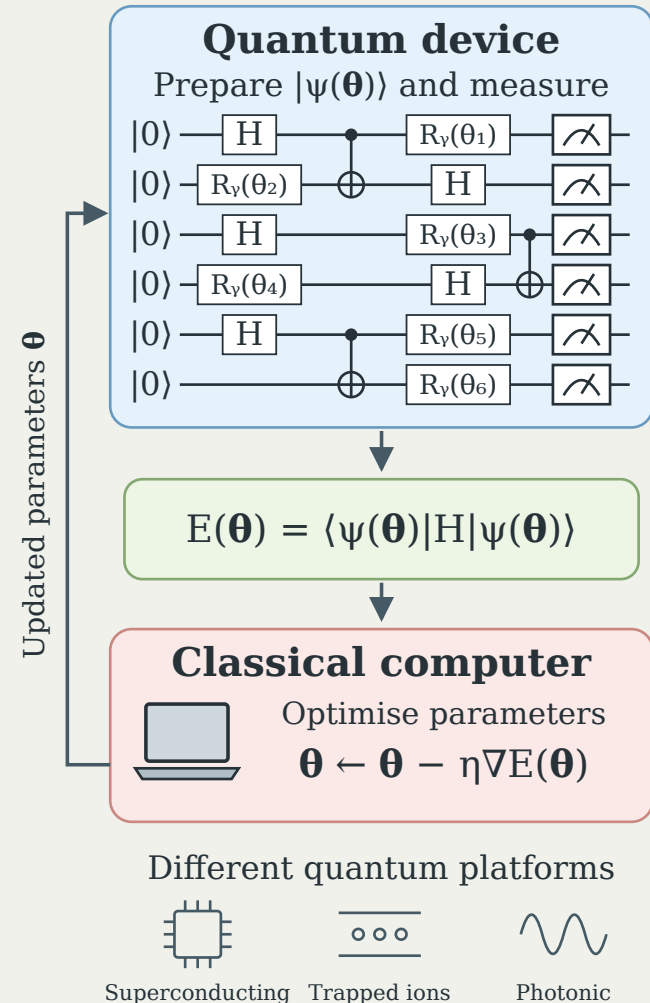
$$E(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle$$

3. Evaluate

Estimate the objective using quantum measurements.

4. Optimise and repeat

Update the parameters on a classical computer.



A general strategy

Neural-network training

Ansatz

Neural network

Objective

Minimise a loss function

Evaluation

Forward pass and loss calculation

Optimisation

Gradient-based updates

1D quantum ground states

Ansatz

Matrix product state — MPS

Objective

Minimise the energy

Evaluation

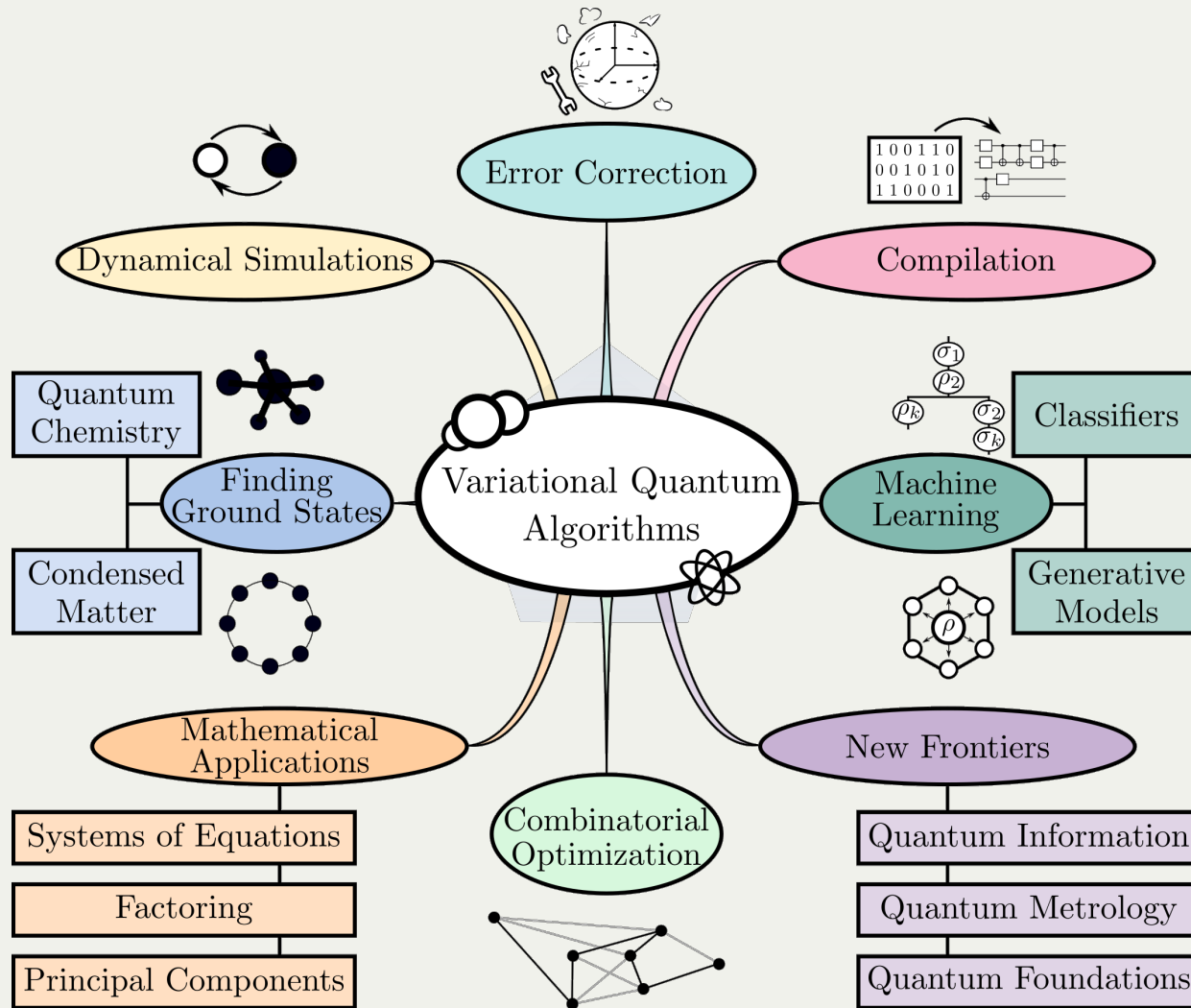
Tensor contractions

Optimisation

DMRG sweeps: local tensor updates
using iterative eigensolvers

DMRG: density-matrix renormalisation group.
A standard classical method for strongly interacting 1D systems.

Applicability



Ansatz preliminaries

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Hermitian: $I = I^\dagger, \quad X = X^\dagger, \quad Y = Y^\dagger, \quad Z = Z^\dagger$

Anticommuting: $XY = -YX, \quad YZ = -ZY, \quad ZX = -XZ$

Every unitary is a complex exponential of a Hermitian matrix:

$$U = e^{-iA}, \quad A = A^\dagger.$$

$$R_x(\theta) = e^{-i\theta X/2}$$

$$\begin{aligned} &= \left[1 - \frac{(\theta/2)^2}{2!} + \frac{(\theta/2)^4}{4!} - \dots \right] I - i \left[\frac{\theta}{2} - \frac{(\theta/2)^3}{3!} + \frac{(\theta/2)^5}{5!} - \dots \right] X \\ &= \cos(\theta/2)I - i \sin(\theta/2)X \end{aligned}$$

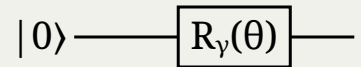
Note: $X^2 = Y^2 = Z^2 = I$.

Rotations and controlled operations

$$R_p(\theta) = e^{-i\theta P/2} = \cos(\theta/2)I - i \sin(\theta/2)P, \quad P \in \{X, Y, Z\}$$

Example:

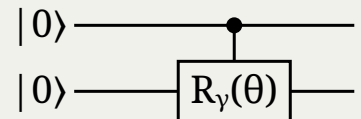
$$R_y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$$



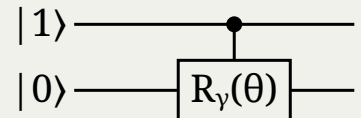
Controlled operation:

$$C(U) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$$

$$CR_y(\theta)|00\rangle = |00\rangle$$



$$CR_y(\theta)|10\rangle = \cos(\theta/2)|10\rangle + \sin(\theta/2)|11\rangle$$

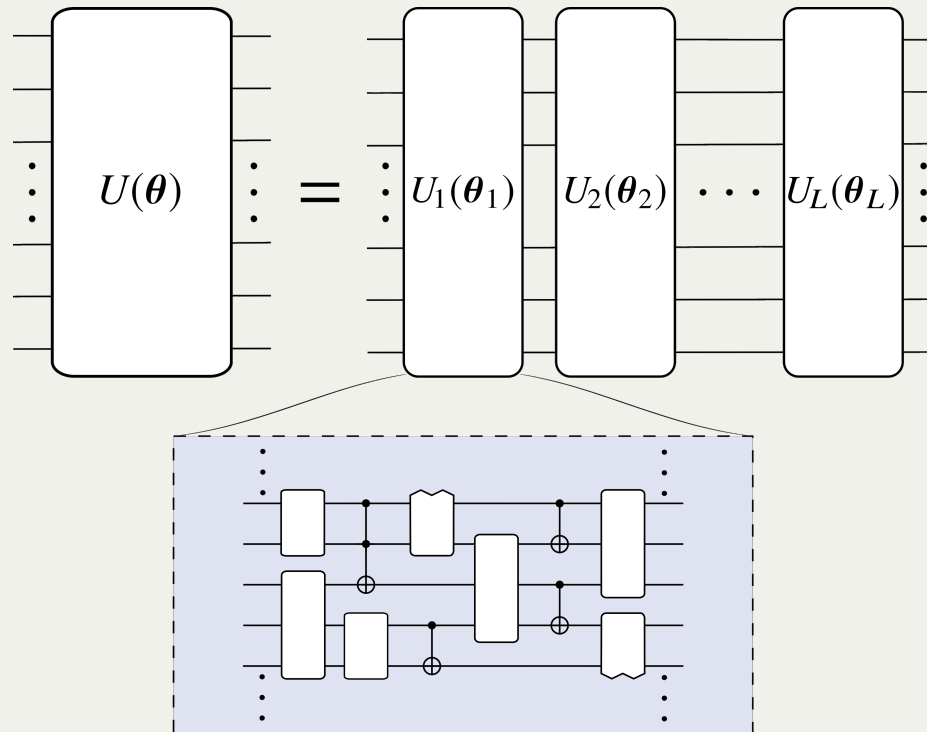


$\theta \in \mathbb{R}$. In $CR_y(\theta) = C(R_y(\theta))$, the first qubit / upper wire is the control.

Generic Ansatz Structure

$$U(\boldsymbol{\theta}) = U_L(\boldsymbol{\theta}_L) \cdots U_1(\boldsymbol{\theta}_1)$$

$$U_\ell(\boldsymbol{\theta}_\ell) = \prod_{j=1}^m [e^{-i\theta_{\ell j} H_j} W_j]$$



Choosing an ansatz

Balance expressivity, trainability and hardware constraints.

- **Hardware-efficient ansatz**

Use native gates and device connectivity to reduce circuit depth.

- **Unitary coupled-cluster ansatz**

Use excitation-based unitaries to capture electronic correlations.

- **Quantum alternating operator ansatz**

Alternate cost-encoding and mixing unitaries.

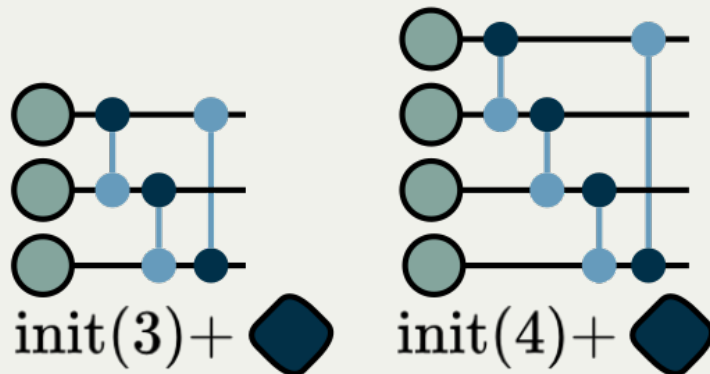
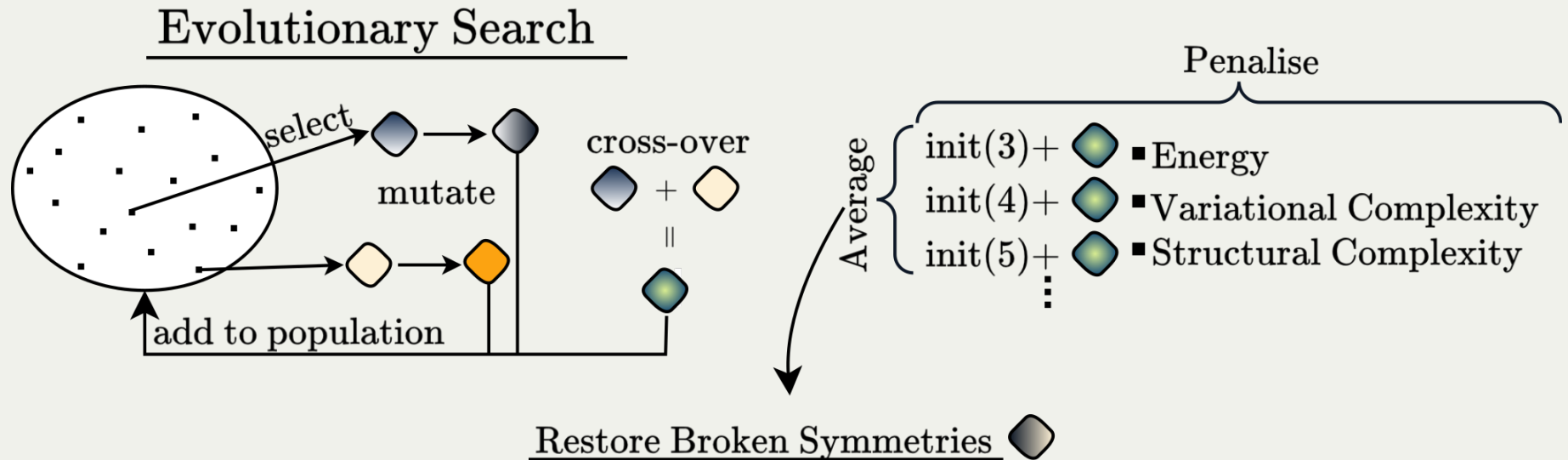
- **Variational Hamiltonian ansatz**

Build layers from parameterised evolutions under Hamiltonian terms.

- **Variable-structure ansatz**

Optimise the circuit structure as well as its parameters.

Variable-structure Ansätze



Primitive Operations: $e^{-i\theta A \otimes B}$ where
 $A, B \in \{I, \frac{1}{2}X, \frac{1}{2}Y, \frac{1}{2}Z\}$

M. Lourens, I. Sinayskiy, J. N. Kriel, and F. Petruccione, PRL, 135, 210401 (2025).

Optimisation

Gradient descent

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta_k \nabla_{\boldsymbol{\theta}} C(\boldsymbol{\theta}_k)$$

- **Adam:** uses gradient averages and squared-gradient averages to adapt step sizes.
- **BFGS / L-BFGS:** uses gradient changes to approximate curvature.
- **Quantum natural gradient:** uses gradients and the geometry of quantum states.
- **SPSA:** estimates the full gradient from two randomly perturbed cost evaluations.
- **COBYLA / Nelder–Mead:** uses cost values without requiring supplied gradients.
- How do we get gradients on a quantum computer?

$\eta_k > 0$: learning rate; k : optimisation step.

Application: finding ground states

1. Ansatz

Normalised trial state $|\psi(\boldsymbol{\theta})\rangle$.

2. Objective

Minimise the energy $\langle\psi(\boldsymbol{\theta})|H|\psi(\boldsymbol{\theta})\rangle$.

3. Optimise

Update $\boldsymbol{\theta}$ to lower the energy.

Why does this work? The variational principle

Variational quantum eigensolver

Goal: estimate the ground-state energy E_0 of H .

Known Hamiltonian

$$H = \sum_{k=1}^M c_k \sigma_k, \quad c_k \in \mathbb{R}$$

Pauli strings σ_k ; the number of terms grows polynomially with the number of qubits.

Preparable trial state

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|\psi_0\rangle$$

Normalised reference state; an efficiently implementable ansatz.

Energy objective

$$C(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle \geq E_0$$

Quantum: measure Pauli strings

$$C(\boldsymbol{\theta}) = \sum_{k=1}^M c_k \langle \sigma_k \rangle_{\boldsymbol{\theta}}$$

Classical: optimise and repeat

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} C(\boldsymbol{\theta})$$

Side note: Linear regression

Same pipeline: choose an ansatz, define an error, then choose the coefficients that make the error as small as possible.

Ansatz

$$|\hat{\mathbf{y}}\rangle = X\mathbf{b} = \sum_j b_j |x_j\rangle$$

Error and objective

$$|\varepsilon\rangle = X\mathbf{b} - |\mathbf{y}\rangle, \quad C(\mathbf{b}) = \frac{1}{2} \langle \varepsilon | \varepsilon \rangle$$

Optimise: the remaining error has no component along any column of X .

$$0 = \frac{\partial C}{\partial b_k} = \frac{1}{2} (\langle x_k | \varepsilon \rangle + \langle \varepsilon | x_k \rangle)$$

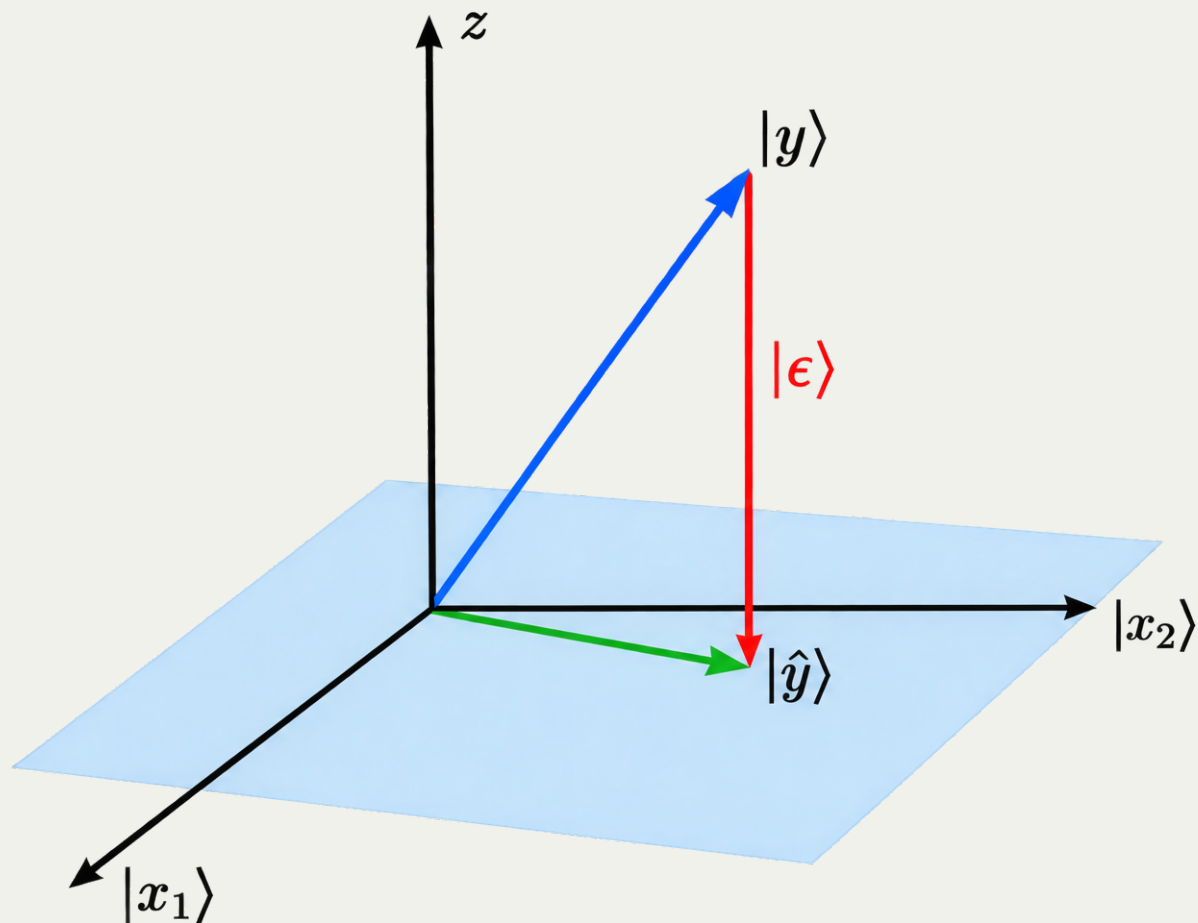
$$\text{real vectors} \quad \implies \quad \langle x_k | \varepsilon \rangle = 0 \quad \forall k$$

$$X^T (X\mathbf{b} - |\mathbf{y}\rangle) = \mathbf{0} \quad \implies \quad X^T X\mathbf{b} = X^T |\mathbf{y}\rangle$$

$$|\hat{\mathbf{y}}\rangle = \underbrace{X(X^T X)^{-1} X^T}_{\Pi} |\mathbf{y}\rangle$$

So linear regression is a projection: the target is replaced by the closest vector reachable by the ansatz.

Linear regression: visual intuition



Application: Quantum Dynamics

$$\frac{d}{dt}|\psi(t)\rangle = -iH|\psi(t)\rangle$$

Ansatz:

$$|\phi(\boldsymbol{\theta}(t))\rangle = U(\boldsymbol{\theta}(t))|\psi_0\rangle \approx |\psi(t)\rangle$$

Target:

$$|y\rangle = -iH|\phi\rangle$$

Ansatz time derivative:

$$\frac{d}{dt}|\phi(\boldsymbol{\theta}(t))\rangle = \sum_j \dot{\theta}_j \frac{\partial U(\boldsymbol{\theta})}{\partial \theta_j} |\psi_0\rangle$$

$$|\partial_j \phi\rangle \equiv \frac{\partial U(\boldsymbol{\theta})}{\partial \theta_j} |\psi_0\rangle$$

$$|\hat{y}\rangle = D\dot{\boldsymbol{\theta}} = \sum_j \dot{\theta}_j |\partial_j \phi\rangle$$

$$D = (|\partial_1 \phi\rangle, \dots, |\partial_p \phi\rangle)$$

Error and objective

$$|\varepsilon\rangle = |\hat{y}\rangle - |y\rangle = D\dot{\boldsymbol{\theta}} + iH|\phi\rangle, \quad C = \frac{1}{2} \langle \varepsilon | \varepsilon \rangle$$

$$\begin{aligned} 0 &= \frac{\partial C}{\partial \dot{\theta}_k} = \frac{1}{2} (\langle \partial_k \phi | \varepsilon \rangle + \langle \varepsilon | \partial_k \phi \rangle) \\ \implies \operatorname{Re} \langle \partial_k \phi | \varepsilon \rangle &= 0 \quad \forall k \end{aligned}$$

Dynamical quantum simulation: solve, step, repeat

$$\operatorname{Re}\left(D^\dagger(D\dot{\boldsymbol{\theta}} - |y\rangle)\right) = \mathbf{0}$$

$$A\dot{\boldsymbol{\theta}} = \mathbf{q}$$

$$A = \operatorname{Re}(D^\dagger D), \quad \mathbf{q} = \operatorname{Re}(D^\dagger |y\rangle)$$

Linear regression

$$X^T X \mathbf{b} = X^T |y\rangle$$

Quantum simulation

$$\operatorname{Re}(D^\dagger D)\dot{\boldsymbol{\theta}} = \operatorname{Re}(D^\dagger |y\rangle)$$

Update parameters and repeat

$$\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}_n + \Delta t \dot{\boldsymbol{\theta}}_n$$

From linear regression to neural networks

A linear model

$$f_b(\mathbf{x}) = \mathbf{b}^T \mathbf{x}$$

Generalise the ansatz: compose layers and add nonlinearities

$$\mathbf{a}_0 = \mathbf{x}, \quad \mathbf{a}_\ell = \sigma(W_\ell \mathbf{a}_{\ell-1} + \mathbf{c}_\ell)$$

$$f_\theta(\mathbf{x}) = \mathbf{w}^T \mathbf{a}_L + c$$

θ collects all weights and biases; e.g. $\sigma(z) = \max(0, z)$, applied componentwise.

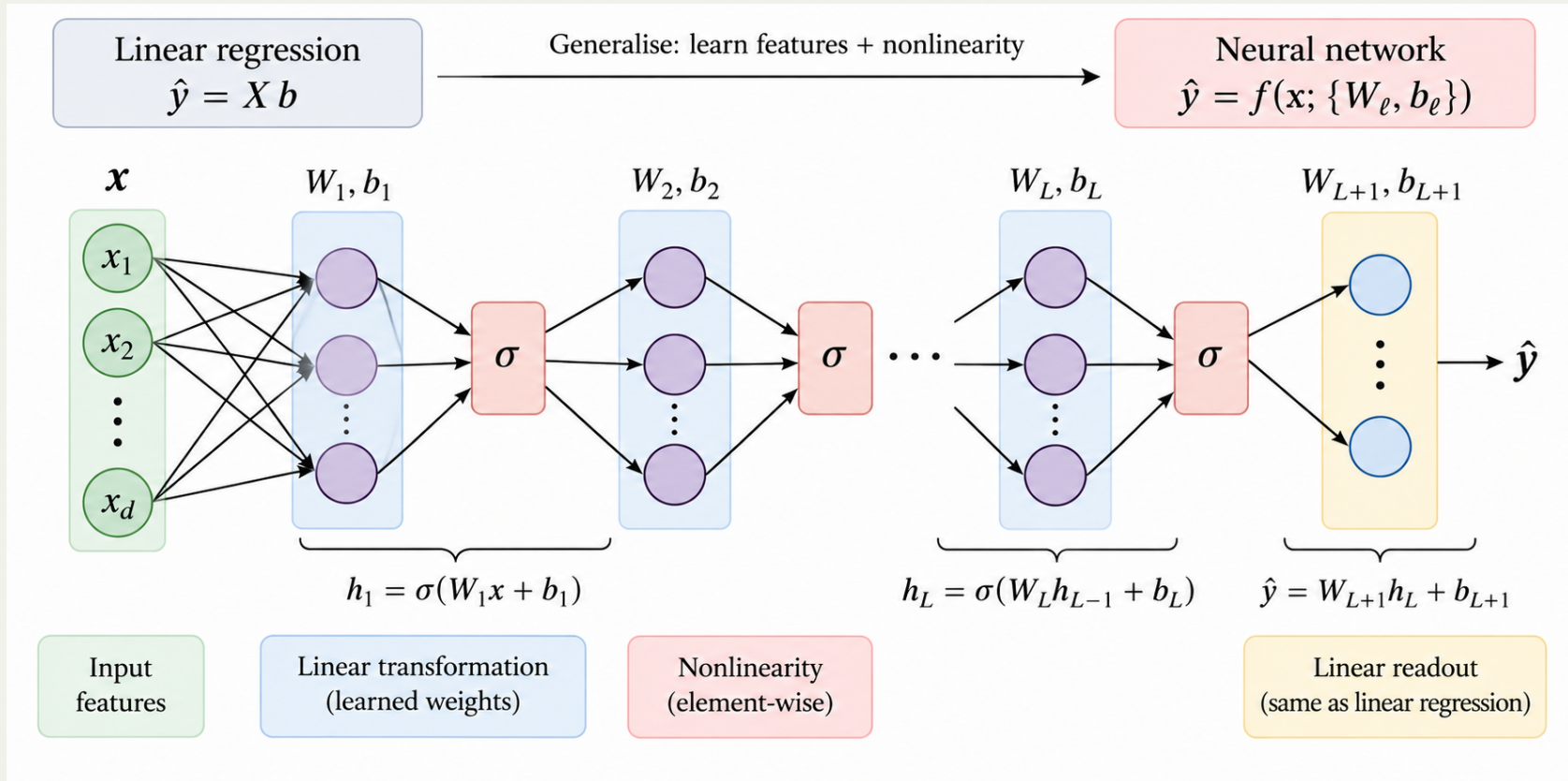
Without nonlinear activations, stacked affine layers are still one affine map.

Keep the objective: fit the training data

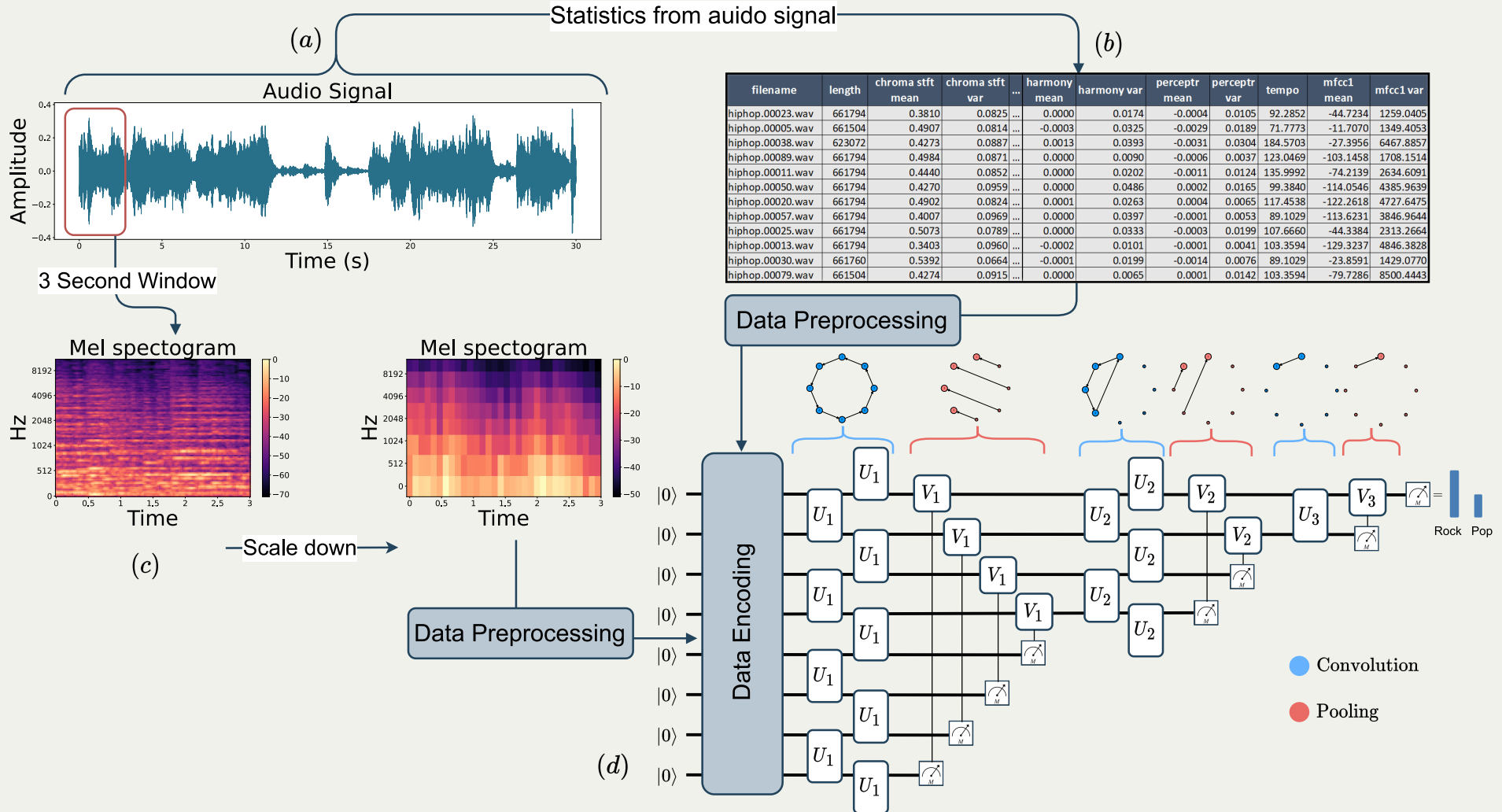
$$C(\theta) = \frac{1}{2N} \sum_{i=1}^N [f_\theta(\mathbf{x}_i) - y_i]^2$$

Backpropagation computes gradients; an optimiser updates parameters.

Neural Networks



Quantum machine learning



VQLS: fit a linear-system solution state

$$X|b\rangle = |y\rangle, \quad |b(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|0\rangle^{\otimes n}$$

Tune $\boldsymbol{\theta}$ so that $X|b(\boldsymbol{\theta})\rangle$ points along $|y\rangle$.

Predicted right-hand side

$$|\hat{y}\rangle = X|b(\boldsymbol{\theta})\rangle$$

Build projector orthogonal to $|y\rangle$

$$P = I - |y\rangle\langle y|$$
$$P^2 = P, \quad P|y\rangle = 0$$

Error: the part of $|\hat{y}\rangle$ orthogonal to $|y\rangle$

$$|\varepsilon\rangle = P|\hat{y}\rangle = (I - |y\rangle\langle y|)X|b(\boldsymbol{\theta})\rangle$$

$$C(\boldsymbol{\theta}) = \frac{\langle \hat{y} | P | \hat{y} \rangle}{\langle \hat{y} | \hat{y} \rangle} = \frac{\langle \varepsilon | \varepsilon \rangle}{\langle \hat{y} | \hat{y} \rangle}$$

$|\hat{y}\rangle$ is not generally normalised because X is not generally unitary.

VQLS: optimise, then use the solution state

$$\begin{aligned} C(\boldsymbol{\theta}) = 0 &\iff X|b(\boldsymbol{\theta})\rangle \propto |y\rangle \\ \implies |b(\boldsymbol{\theta}^*)\rangle &\approx \frac{X^{-1}|y\rangle}{\|X^{-1}|y\rangle\|} \end{aligned}$$

Quantum: estimate $C(\boldsymbol{\theta})$.

Classical: update $\boldsymbol{\theta}$ to reduce the orthogonal component.

$$\boldsymbol{\theta} \longrightarrow |b(\boldsymbol{\theta})\rangle \longrightarrow |\hat{y}\rangle = X|b(\boldsymbol{\theta})\rangle \longrightarrow C(\boldsymbol{\theta}) \longrightarrow \boldsymbol{\theta}_{\text{new}}$$

Use case

$$\langle O \rangle = \langle b(\boldsymbol{\theta}^*) | O | b(\boldsymbol{\theta}^*) \rangle$$

We use the prepared solution state directly, rather than reading out every amplitude.

If $|y\rangle$ is not normalised, use $P = I - \frac{|y\rangle\langle y|}{\langle y|y\rangle}$.

QAOA by example: MaxCut on a triangle

Problem: choose bits $z_i \in \{0, 1\}$. An edge is good if its endpoints have different bits.

Cost Hamiltonian: penalise uncut edges

$$H_C = \sum_{(i,j) \in E} \frac{I + Z_i Z_j}{2}$$

Low energy means many edges are cut. For a triangle, the best cost is 1.

Ansatz: alternate scoring and mixing

$$|\psi(\gamma, \beta)\rangle = \prod_{\ell=1}^p e^{-i\beta_\ell H_M} e^{-i\gamma_\ell H_C} |+\rangle^{\otimes 3}, \quad H_M = -\sum_i X_i$$

Objective

$$\min_{\gamma, \beta} \langle H_C \rangle$$

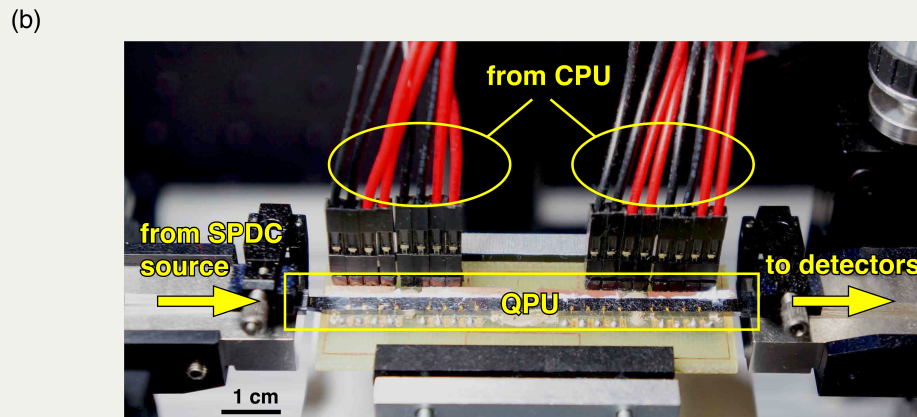
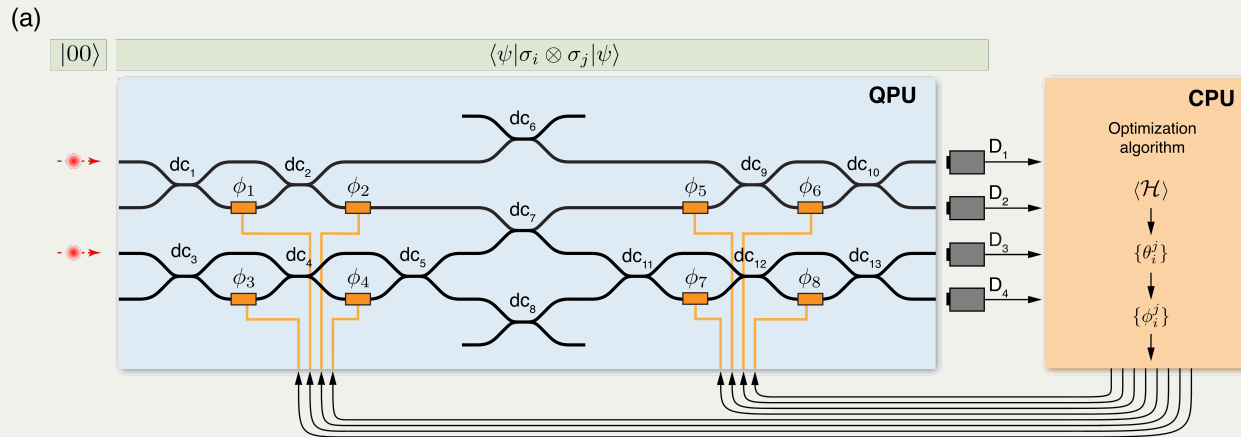
Tune the angles so measurement samples are biased toward good cuts.

Output: measure bitstrings such as 001, 010, 100; keep the lowest-cost samples.

Typical QAOA problems: MaxCut, graph partitioning, constraint satisfaction, scheduling, routing.

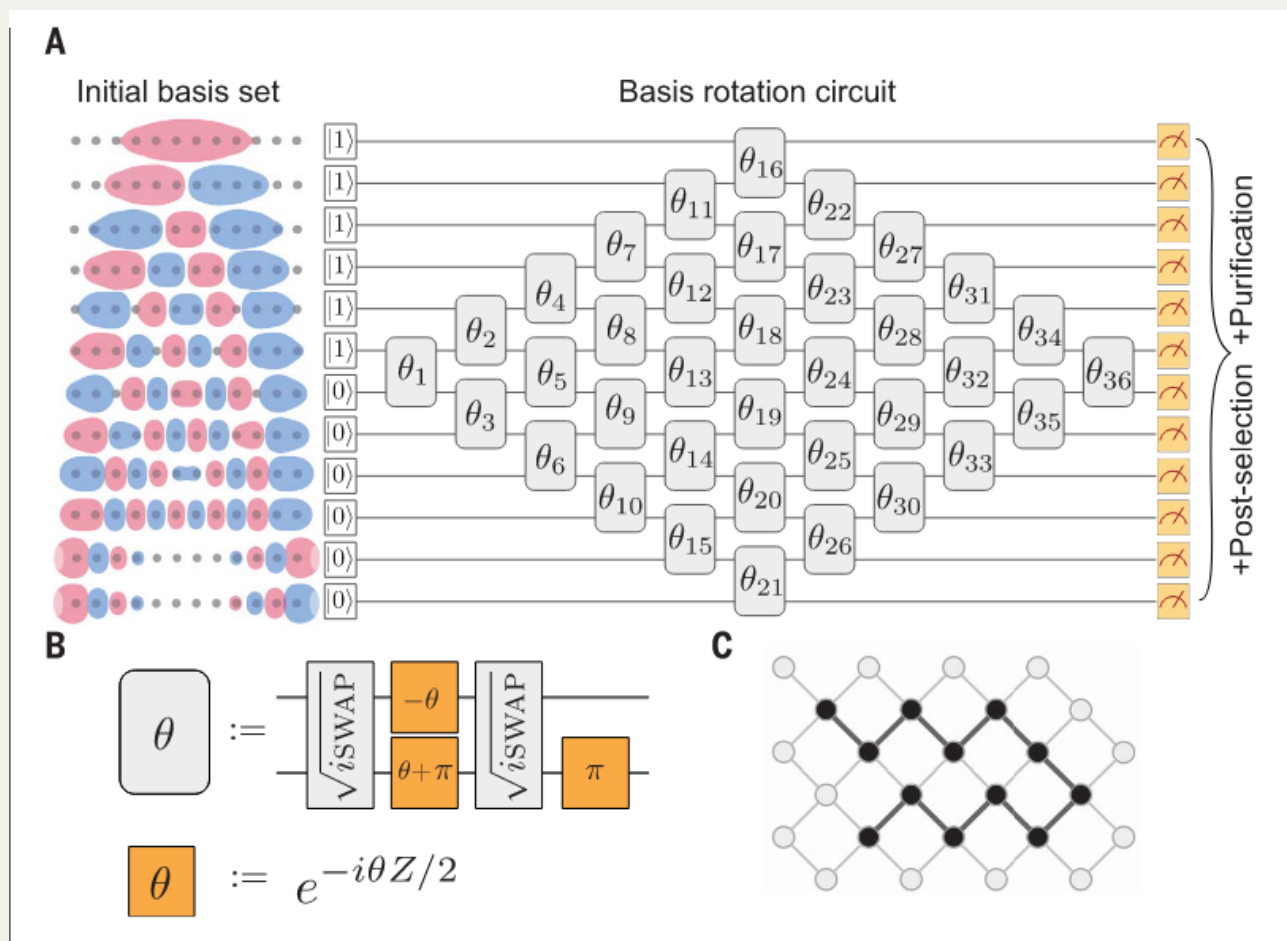
Finding the ground state of a helium hydride ion

First experimental demonstration: ground-state energy and bond-dissociation curve of HeH^+ on a two-qubit photonic quantum processor.



Electronic structure of hydrogen chains

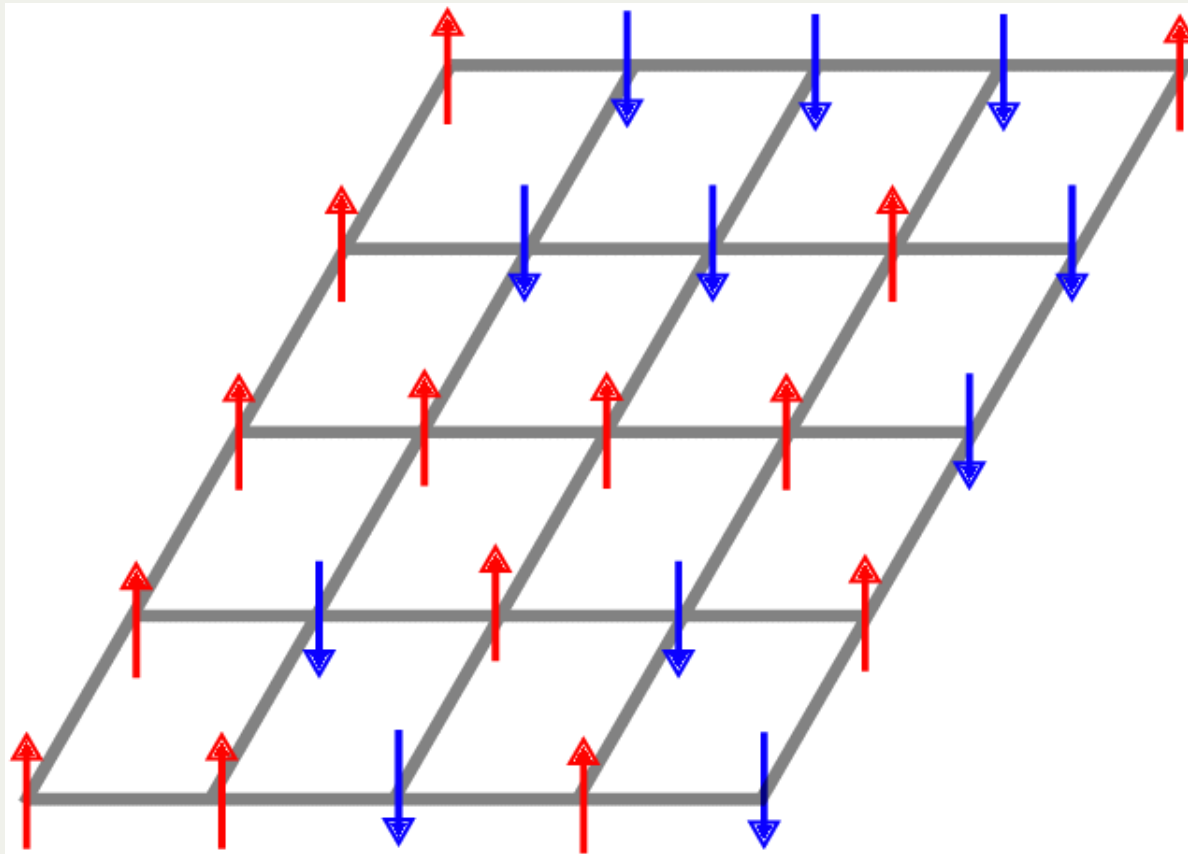
Hartree–Fock ground-state energies of hydrogen chains up to H_{12} on Google's Sycamore superconducting processor.



Google AI Quantum and Collaborators, Science **369**, 1084–1089 (2020).

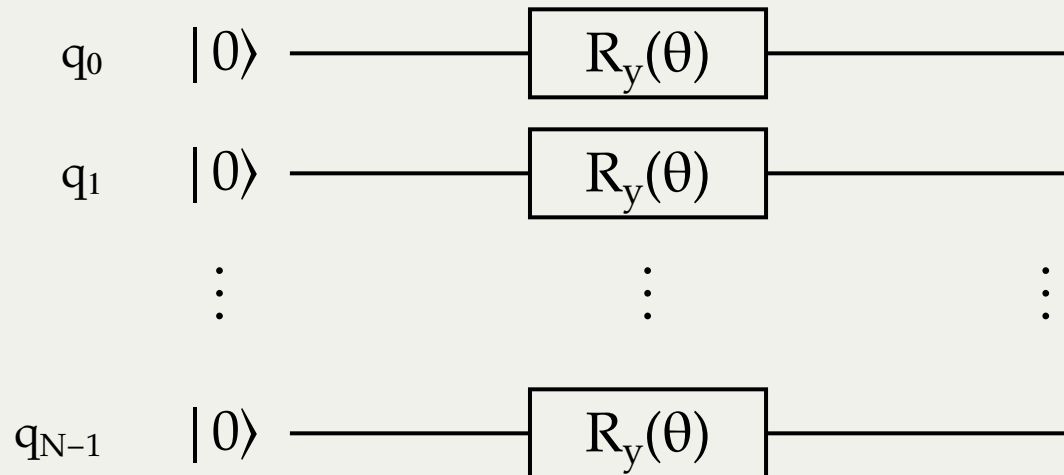
Transverse-field Ising model

$$H = -J \sum_{\langle i,j \rangle} Z_i Z_j - h \sum_i X_i$$



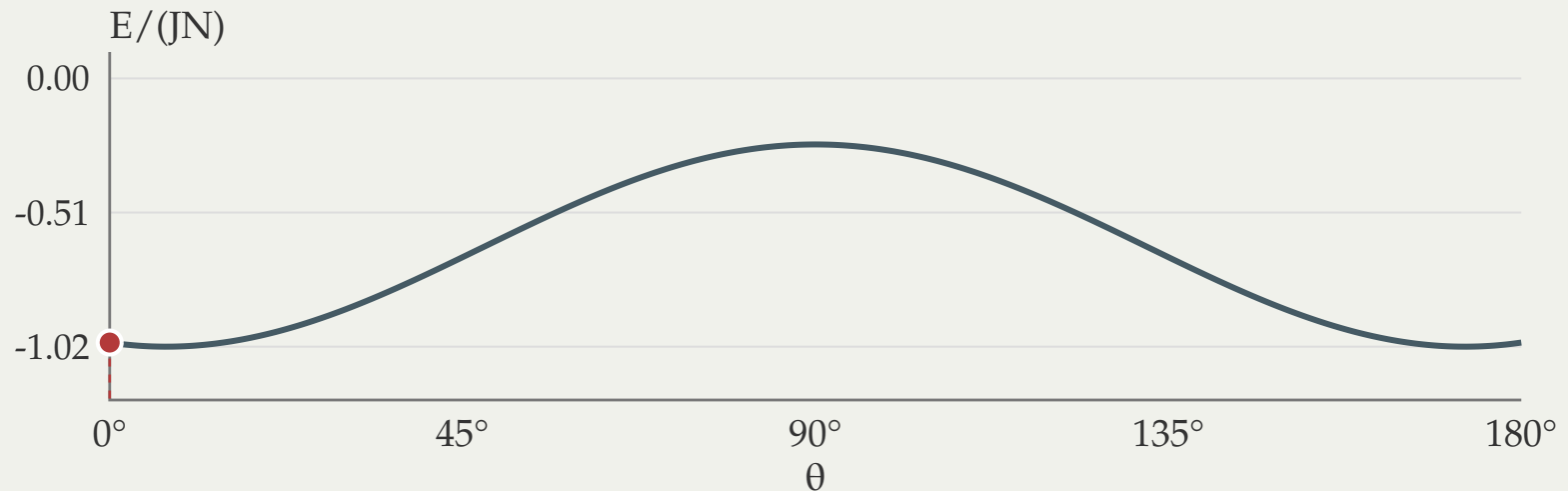
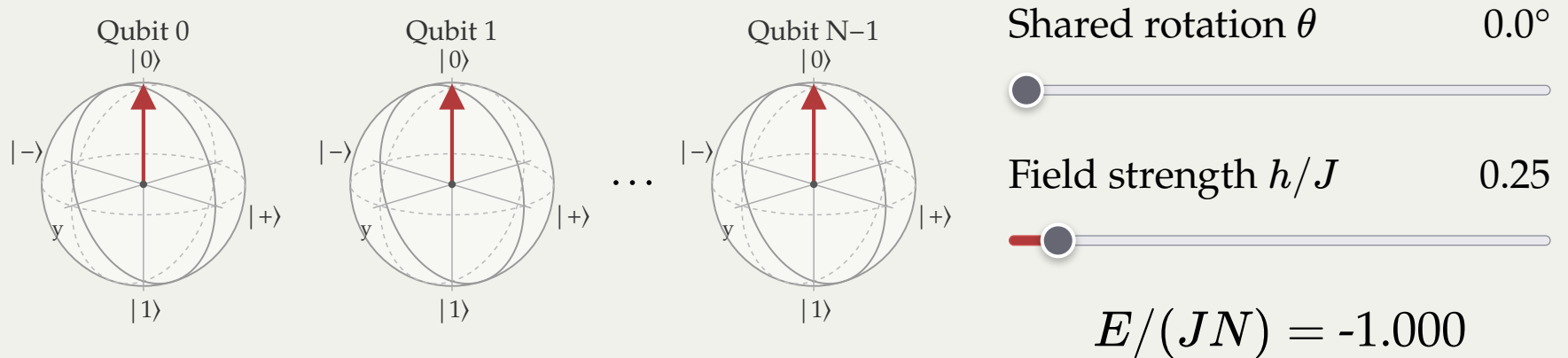
A product-state ansatz

$$|\psi(\theta)\rangle = \left[\cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} |1\rangle \right]^{\otimes N}$$



Same θ on every qubit; no entangling gates — a uniform mean-field ansatz.

Model qualitative overview



$|0 \dots 0\rangle$

$|+ \dots +\rangle$

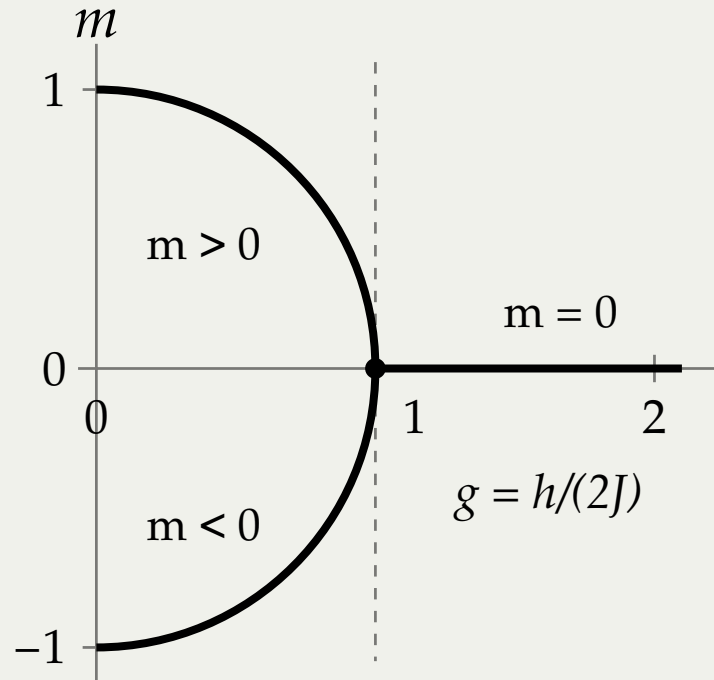
$|1 \dots 1\rangle$

Minimise

Uniform product-state (mean-field) approximation; “Minimise” searches only this ansatz.

The product ansatz and mean-field theory

$$m = \frac{1}{N} \sum_{i=0}^{N-1} \langle Z_i \rangle = \cos \theta, \quad g = \frac{h}{2J}$$



Two symmetry-related minima for $g < 1$

$$\boxed{m^2 + g^2 = 1} \quad (0 \leq g \leq 1)$$

Mean-field critical field: $h_c^{\text{MF}} = 2J$.

$T = 0, J > 0$, periodic chain. The exact infinite-chain critical field is $h_c = J$.

Pfeuty, Ann. Phys. 57, 79–90 (1970).